46755 - Renewables in Electricity Markets

Assignment 2: Market Participation (Stakeholder Perspective)

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This assignment contains two steps:

> **Step 1) Participation in Day-ahead and Balancing Markets** is based on Lectures 8 and 9 and consists of 4 tasks (1.1–1.4).
>
> **Step 2) Participation in Ancillary Service Markets** is based on Lecture 10 and consists of 3 tasks (2.1–2.3).

Note: Please refer to the introduction slides presented on February 3, 2025, for the report template and page limit. A grade deduction of 1.5% will be applied for each page exceeding the limit, except for the appendix, which should only include input data.

Step 1) Participation in Day-ahead and Balancing Markets

Problem Description and Scenario Generation

Consider a price-taking wind farm with an installed capacity of 500 MW. The objective is to formulate and solve an optimization problem to determine its optimal offering strategy in terms of production quantity in the day-ahead market. The offer price is assumed to be zero. The analysis spans a 24-hour period, focusing exclusively on the day-ahead and balancing markets while excluding the reserve and intraday markets.

For each hour of the following day, the analysis considers **three independent sources of uncertainty:**

1. Wind power production,
2. Day-ahead market price,
3. The real-time power system imbalance (whether the system experiences a supply deficit or excess).

To model these uncertainties, construct at least **1,600 combined scenarios** representing their joint realizations. Because the three sources are assumed to be *mutually uncorrelated*, you must generate scenarios for each source independently. The total number of combined scenarios is then the *product* of the number of scenarios generated for each source.

For example, if you generate:

20 wind scenarios, 20 price scenarios, 4 imbalance scenarios,

then the total number of joint scenarios is:

$$20 \times 20 \times 4 = 1{,}600.$$

Note: You are free to use different numbers of scenarios for each source of uncertainty, as long as the total number of combined scenarios $\geq 1{,}600$.

Note: Historical data can be used for scenario generation, where each realization from a past day may be considered one scenario.

1

Scenario Generation for Wind Power Production Forecast

Realistic wind power production scenarios can be generated using:

- FINGRID (Finnish TSO)
- ELIA (Belgian TSO)
- DataSMOPF wind scenarios
- Renewables.ninja

2

Scenario Generation for Day-Ahead Price Forecast

Use historical hourly day-ahead prices from the Nord Pool market, assuming the wind farm is located in DK2:

- Nord Pool day-ahead prices (DK2)

3

Scenario Generation for Real-Time System Imbalance

To model the real-time system imbalance (SI), generate 24 binary random variables (one per hour), representing either:

- supply deficit (SI = 1), or
- supply surplus (SI = 0).

A Bernoulli distribution or an empirical distribution may be used.

Scenario Generation for Balancing Price Forecasts

For simplicity, balancing prices can be modeled as a scaled version of the day-ahead price in the corresponding scenario:

- If the system is in deficit: BP $= 1.25 \times$ DA price,
- If the system is in surplus: BP $= 0.85 \times$ DA price.

Reminder (Lectures 5b and 8):

1. Under the **one-price scheme**, the wind farm always settles imbalances at the balancing price.
2. Under the **two-price scheme**, the settlement price depends on whether the imbalance caused by the wind farm is beneficial or harmful to the system. This must be determined within the optimization.

Task 1.1) Offering Strategy Under a One-Price Balancing Scheme

Following Lecture 8, formulate and solve the stochastic offering problem under the one-price scheme using the in-sample scenarios. Determine the optimal hourly day-ahead offers and compute the expected profit. Also illustrate the profit distribution across scenarios. Discuss whether the strategy exhibits all-or-nothing bidding behavior, and explain why.

Task 1.2) Offering Strategy Under a Two-Price Balancing Scheme

Repeat Task 1.1 under the two-price scheme. Compare both offering strategies and profit distributions, and explain key differences.

Task 1.3) Ex-post Analysis

Using 200 in-sample and 1,400 out-of-sample scenarios, perform an 8-fold cross-validation (as discussed in Lecture 8). For each run, compute expected profits using both in-sample and out-of-sample scenarios. After all 8 runs, compare averaged in-sample vs. out-of-sample expected profits. Discuss whether the offering decisions generalize well and whether modifying the in-sample size (while keeping total scenarios at 1,600) could improve decision quality.

Task 1.4) Risk-Averse Offering Strategy

Following Lecture 9, formulate and solve the risk-averse offering strategy under both one- and two-price schemes, using CVaR with $\alpha = 0.90$. Gradually increase β from 0 upwards and plot expected profit versus CVaR. Discuss how the offering strategy and profit variability evolve with risk aversion. Explain how the profit distribution changes and assess whether altering the in-sample scenarios significantly affects the risk-averse solutions.

Step 2) Participation in Ancillary Service Markets

Problem Description and Scenario Generation

In Lecture 10, we explore the offering strategy for a price-taking stochastic flexible load with consumption between 0 and 600 kW. We focus on the DK2 **FCR-D UP** market with hourly bids. The flexible load can provide upward frequency containment by reducing consumption quickly.

We consider a single bidding hour, modeled with minute-level resolution.

Data Generation for Future Stochastic Load

Randomly generate 300 load profiles such that:

- consumption remains between 220 and 600 kW,
- minute-to-minute changes do not exceed 35 kW.

From these profiles:

- 100 are used as **in-sample** profiles (each equally likely),
- 200 are used as **out-of-sample** profiles.

Task 2.1) In-sample Decision Making: Offering Strategy Under the P90 Requirement

Using the 100 in-sample profiles, determine the optimal FCR-D UP reserve bid (in kW) satisfying Energinet's P90 requirement. Solve the problem using both ALSO-X and CVaR.

Task 2.2) Verification of the P90 Requirement Using Out-of-Sample Analysis

Using the 200 out-of-sample profiles, verify whether the P90 requirement is met for each solution method. No optimization is needed—compare the chosen reserve bid with the minute-level consumption profiles to evaluate possible shortfalls.

Task 2.3) Energinet Perspective

Analyze how modifying the P90 requirement (e.g., threshold between 80% and 100%) affects the optimal reserve bid (in-sample) and the expected reserve shortfall (out-of-sample) when using ALSO-X. Discuss whether a trade-off emerges between higher reliability and reduced reserve provision.